

Automated Network Request Management

ServiceNow Project Documentation

Catalog-driven request management with variable mapping, approvals, workflow automation, and controlled task processing

1. Project Overview

The Automated Network Request Management solution is designed to capture network-related requests through a ServiceNow catalog experience and process them through a controlled workflow. The solution uses structured request information, variable-to-field mapping, approval rules, and task automation to provide a consistent process from request submission through fulfillment.

2. Request Variables

The catalog item collects the information required to understand and process a network request. The primary variables included in the solution are:

- Request type – identifies the type of network service or access being requested.
- Justification – records the business reason for the request and supports approval decisions.
- Portal details – captures the relevant portal or access information associated with the request.
- Urgency – indicates how quickly the request needs to be addressed and can be used in prioritization.

3. Approval Use Cases

Approval routing is based on the nature and sensitivity of the request:

- Manager approval for standard requests – routine requests are routed to the appropriate manager for authorization.
- Network security approval for high-sensitivity requests – requests involving higher security sensitivity are routed for additional security review.
- Group approval for department-specific tasks – requests that belong to a specific department or operational group can be routed to the responsible group for approval.

4. Solution Components Included

Flow Designer workflows: The workflow automates request processing, approval routing, record updates, and task creation. The screenshots below document the implemented flow steps.

Variable-to-field mapping: Catalog variables are mapped into fields on the custom request table so that submitted information is stored as a structured record and can be used by subsequent workflow actions.

Custom table schema: The solution uses the custom tables u_network_database and u_network_task to store network request information and related task information.

Approval condition logic: Flow conditions determine which approval path is required based on request characteristics such as sensitivity, request type, urgency, or department.

Portal widget references: Where portal customizations are used, the portal components provide the customer-facing entry point for submitting and viewing the network request.

5. Flow Designer Workflow – Screenshot Evidence

Figure 1 – Catalog request and workflow entry

This screen represents the starting point of the automated process. The request is initiated through the catalog and the submitted variables become the source data for the workflow.

The screenshot shows the 'Catalog Item' configuration page for 'Network Requester'. The page includes a header with navigation icons and buttons like 'Copy', 'Try It', 'Update', 'Edit in Catalog Builder', and 'Delete'. Below the header, there's a section for 'Enter a Price, approvals, variables, and other information as needed.' The main configuration area contains several fields: 'Name' (Network Requester), 'Application' (Global), 'Active' (checked), 'Fulfillment automation level' (Unspecified), 'Catalogs' (Service Catalog), 'Category' (Network Standard Changes), 'State' (None), 'Checked out' (None), and 'Owner' (System Administrator). Each field has a search icon and a help icon.

Screenshot 1: ServiceNow configuration evidence

Figure 2 – Variable-to-field mapping

The mapping logic transfers catalog variable values into the corresponding fields of the custom network request record. This ensures that information entered by the requester is retained in a structured format.

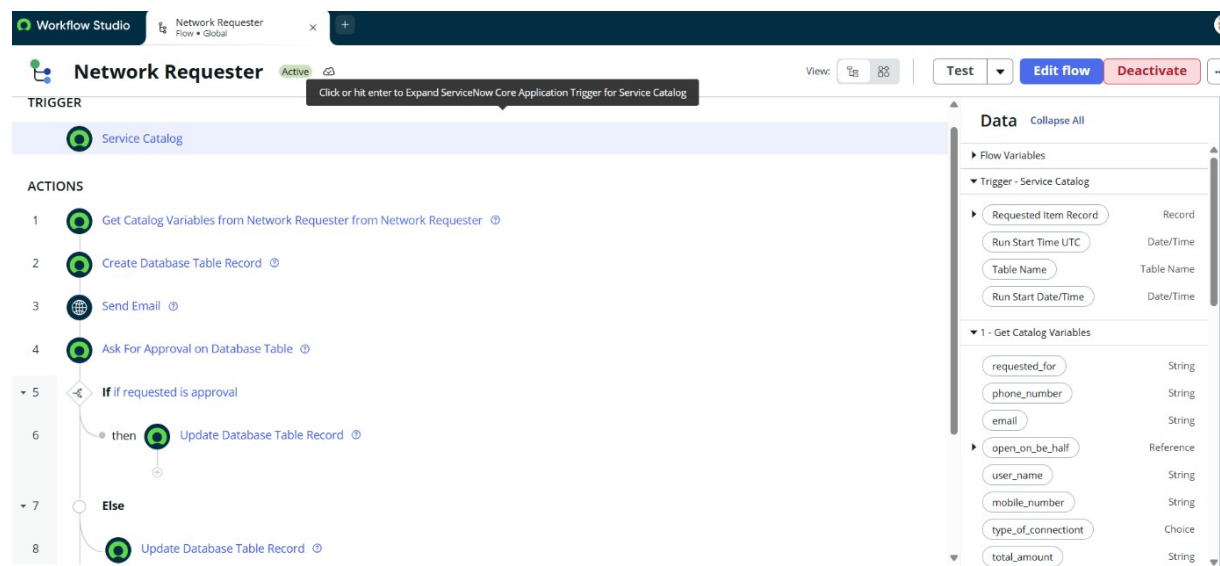
The screenshot shows the 'Variable-to-field mapping' configuration page. It includes a 'Related Links' section with links to 'Item Diagnostic', 'Run Point Scan', and 'SN Utils Versions (32)'. Below this is a navigation bar with tabs for 'Variables (7)', 'Variable Sets (1)', 'Catalog UI Policies (1)', 'Catalog Client Scripts', 'Available For', 'Not Available For', 'Categories (1)', 'Catalogs (1)', 'Catalog Data Lookup Definitions', 'Related Articles', and 'Related Catalog Items'. The 'Variables (7)' tab is selected. The main area shows a table with columns 'Type', 'Question', and 'Order'. The table lists seven variables: 'Multiple Choice' (Type Of Connection, Order 300), 'Single Line Text' (work status, Order 600), 'Single Line Text' (Requested For, Order 100), 'Single Line Text' (Address, Order 700), 'Single Line Text' (Enter Your Existing Id, Order 400), 'Single Line Text' (Mobile Number, Order 200), and 'Single Line Text' (problem, Order 500). The table is paginated, showing '1 to 7 of 7'.

Type	Question	Order
Multiple Choice	Type Of Connection	300
Single Line Text	work status	600
Single Line Text	Requested For	100
Single Line Text	Address	700
Single Line Text	Enter Your Existing Id	400
Single Line Text	Mobile Number	200
Single Line Text	problem	500

Screenshot 2: ServiceNow configuration evidence

Figure 3 – Request record creation

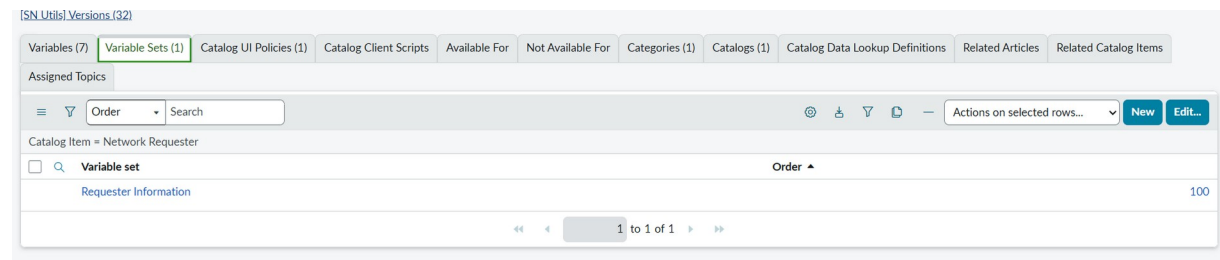
The workflow creates or updates the network request record using the mapped values. The custom record becomes the central record used for approval and fulfillment processing.



Screenshot 3: ServiceNow configuration evidence

Figure 4 – Approval routing

This step demonstrates the approval portion of the workflow. The request is evaluated and routed to the appropriate approver according to the configured approval conditions.



Screenshot 4: ServiceNow configuration evidence

Figure 5 – Approval condition logic

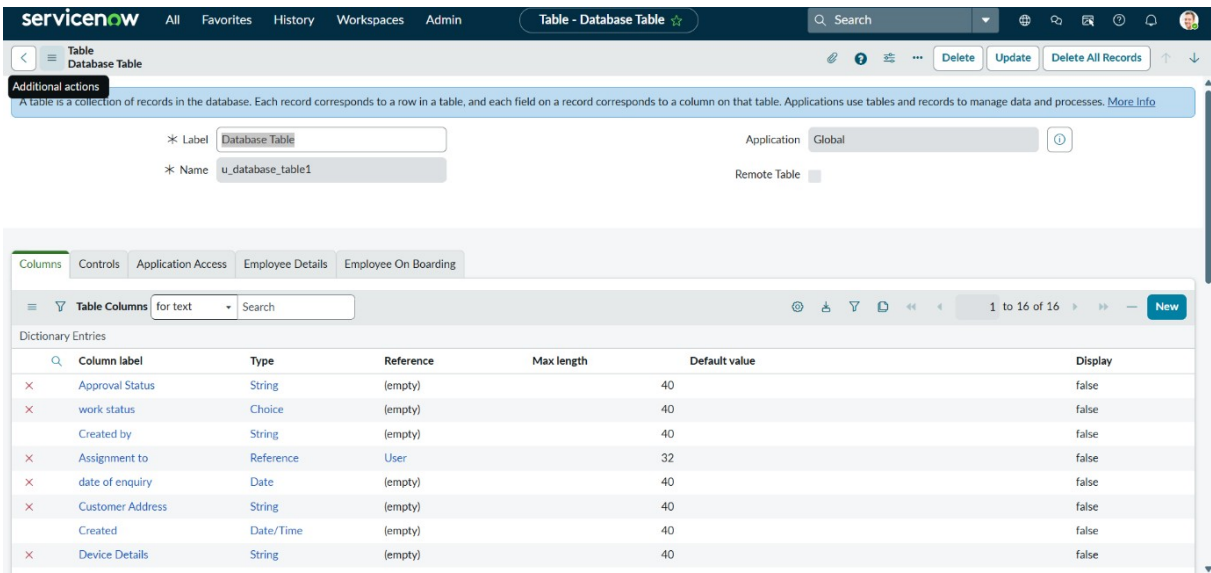
Conditional logic determines the required approval path. Standard requests can follow manager approval, while high-sensitivity or department-specific requests can be routed to additional approval groups.



Screenshot 5: ServiceNow configuration evidence

Figure 6 – Network task creation

After the required approval is completed, the workflow can create a related network task. The task provides the fulfillment team with a clear action to complete.



Screenshot 6: ServiceNow configuration evidence

Figure 7 – Task and request update

The workflow updates the relevant records as processing continues. This maintains visibility of the request and its fulfillment status.

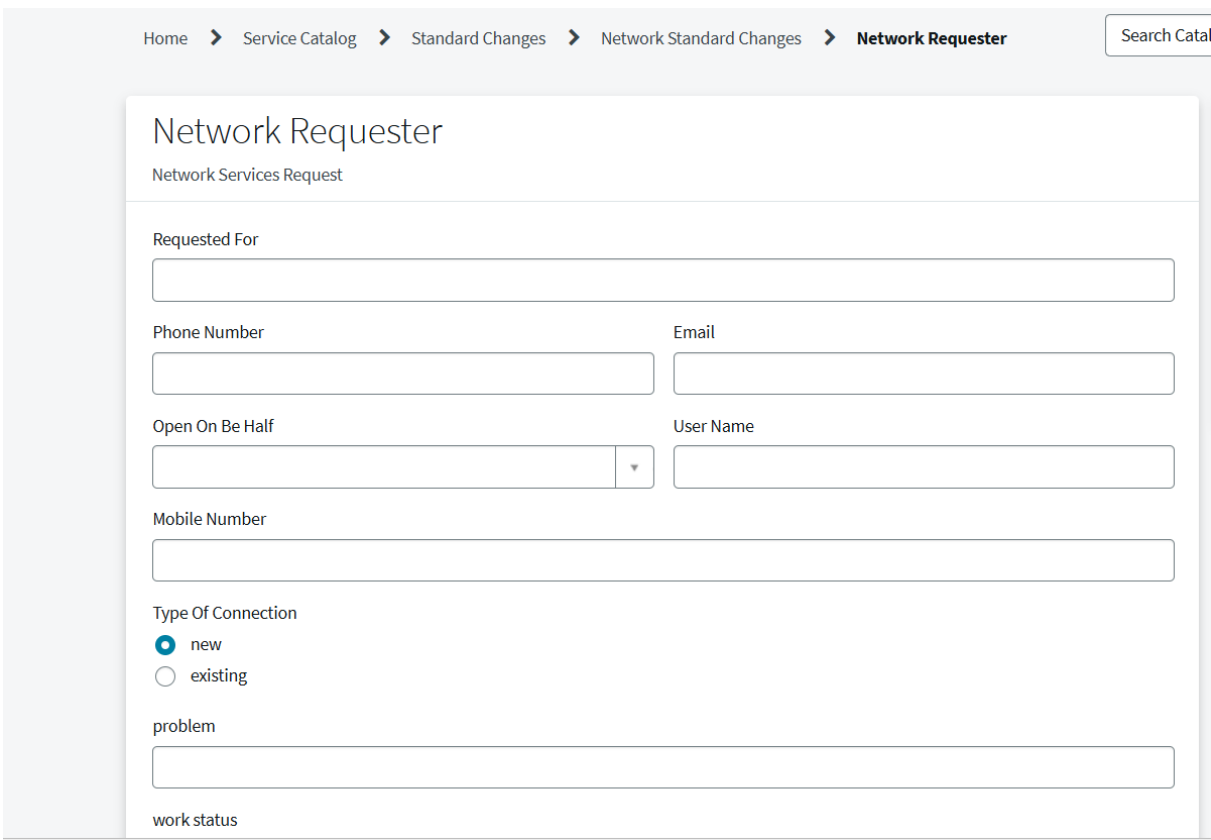


Figure 8 – End-to-end workflow validation

The final screenshot provides evidence that the configured workflow can be reviewed and tested as an end-to-end process, from request submission through approval and task processing.

6. Custom Table Schema

The custom data model separates the main network request information from operational task information. The `u_network_database` table acts as the main request record, while `u_network_task` stores the fulfillment task details related to the request.

Table	Purpose	Workflow Use
<code>u_network_database</code>	Stores the submitted network request and its business/request details.	Source record for mapping, approvals, status tracking, and workflow processing.
<code>u_network_task</code>	Stores operational tasks generated for fulfillment.	Used by the responsible team to complete the required network activity.

7. Variable-to-Field Mapping Logic

When a requester submits the catalog item, the workflow reads the catalog variables and maps them to the appropriate fields in `u_network_database`. This approach prevents important request information from remaining only in the catalog submission and makes the data available for conditions, approvals, reporting, and task creation.

- Request type → corresponding request type field
- Justification → business justification field
- Portal details → portal/access details field
- Urgency → urgency or priority-related field

8. Approval Condition Logic

The workflow evaluates the request before determining the approval route. A standard request follows the manager approval path. If the request requires a higher level of security review, the workflow routes it to network security approval. Department-specific work can be routed to the relevant group for approval. This conditional approach helps apply the correct level of review without using the same approval path for every request.

9. Portal and User Experience

The catalog item provides the requester-facing entry point. The portal experience collects the required variables in a structured manner, while workflow automation handles the backend

processing. If custom portal widgets are used, they can be referenced as part of the presentation layer without changing the core request and workflow data model.

10. Step-by-Step PDI Recreation Manual

- 1. Create the application and custom tables:** Create the application/scope for the solution and create `u_network_database` and `u_network_task` with the fields required by the request and task processes.
- 2. Create the catalog item:** Create a catalog item for network requests and configure the requester-facing information and submission behavior.
- 3. Configure variables:** Add Request type, Justification, Portal details, Urgency, and any other fields required by the implementation.
- 4. Configure variable mapping:** Map each catalog variable to the corresponding field in `u_network_database` so the submitted information becomes part of the structured request record.
- 5. Build the Flow Designer workflow:** Create the trigger and actions needed to create/update the request record, evaluate conditions, request approvals, create fulfillment tasks, and update the request status.
- 6. Configure approval conditions:** Add the decision logic that separates standard manager approval, high-sensitivity network security approval, and department-specific group approval.
- 7. Configure ACLs and roles:** Apply appropriate access controls so requesters, approvers, and fulfillment users can access only the records and actions needed for their responsibilities.
- 8. Test with sample data:** Submit representative requests covering standard, high-sensitivity, and department-specific scenarios. Confirm record creation, mapping, approval routing, task creation, and status updates.
- 9. Validate screenshots:** Capture screenshots in the same order as the workflow steps so another user can reproduce and verify the implementation in a new PDI.

11. Testing Scenarios

Scenario	Expected Route	Expected Result	Validation
Standard request	Manager approval	Request proceeds after manager approval and fulfillment task can be created.	Verify request and task records.
High-sensitivity request	Network security approval	Additional security review is required before fulfillment.	Verify approval condition and approver.
Department-specific task	Relevant group approval	Request is routed to the appropriate	Verify group assignment and task.

		department/group.	
Rejected request	Approval rejection path	Request does not proceed to normal fulfillment.	Verify state/status and approval result.

12. Conclusion

The Automated Network Request Management solution demonstrates how ServiceNow can combine catalog intake, structured data, Flow Designer automation, conditional approvals, custom tables, access control, and fulfillment tasks into a single controlled process. The included screenshots and recreation steps provide implementation evidence and a practical guide for rebuilding and validating the solution in a new Personal Developer Instance.